

SOFTWARE DESIGN DOCUMENT

Open Access Telehealth Platform

Democratizing Access to Healthcare

Version	0.2 — Diagnosis-Permission Model
Status	In Progress — Revised Assumptions
Date	March 28, 2026
Key Change	Assumes government authorization for diagnosis per jurisdiction
Owner	[TBD]
Reviewers	[TBD]

1. Introduction

1.1 Purpose

This Software Design Document (SDD) describes the architecture, modules, data flows, and technical constraints of the Open Access Telehealth Platform. It is a living document updated as decisions are finalized.

1.2 Revised Core Assumption (v0.2)

This version of the SDD operates under the following foundational assumption:

The platform operates with formal government authorization to conduct medical diagnosis in each jurisdiction where it is active.

Physicians on the platform may issue formal diagnoses to patients in jurisdictions where this authorization has been granted. Capabilities that are not yet authorized in a given country — such as prescribing, specialist referral, or lab ordering — remain restricted and are disclosed to both the physician and the patient during every session.

This assumption does not eliminate jurisdictional variation — it reframes it. Rather than a binary 'permitted / not permitted' for diagnosis, the platform now manages a per-country capability matrix that specifies exactly what is and is not authorized. Countries where government authorization has not yet been obtained remain in a 'guidance-only' mode until clearance is granted.

1.3 Scope

The platform enables patients to receive medical diagnosis and clinical care from licensed physicians via three modalities: live video, audio-only call, and asynchronous video submission. The scope of care a physician can deliver is dynamically constrained by their jurisdiction-specific permission profile, which is surfaced to both physician and patient at the start of every session.

1.4 Document Conventions

[TBD]	Open decision — requires resolution before implementation
[Note]	Advisory or constraint requiring stakeholder review
Tier 1–4	Jurisdictional capability tiers — see Section 3

2. Jurisdictional Permission Model

2.1 Overview

The platform's authorization model is built around the concept of a Jurisdictional Permission Tier. Every country in which the platform operates is assigned a tier that defines the maximum scope of clinical action a physician may take when serving a patient in that country. These tiers are set by the Legal and Regulatory Affairs team based on formal government authorization status.

The tier assigned to a patient-jurisdiction pair governs the session — not the physician's home country, and not the physician's personal licensure scope. A physician with broad privileges in their home country is still constrained by the patient's jurisdiction tier.

2.2 Capability Tier Definitions

Tier	Label	Authorized Capabilities	Restrictions
Tier 1	Full diagnosis + treatment + prescribing	Formal diagnosis Treatment recommendation Electronic prescribing (where infrastructure exists) Specialist referral initiation	e-Prescribing requires pharmacy integration. Controlled substances excluded in all tiers.
Tier 2	Full diagnosis + treatment, no prescribing	Formal diagnosis Treatment recommendation Specialist referral initiation No prescribing	Physician must route prescribing needs to local provider or pharmacy partner.
Tier 3	Diagnosis permitted, guidance and referral only	Formal diagnosis Home management guidance Referral recommendation (non-initiating) No treatment prescription No e-prescribing	Government has authorized diagnosis but not treatment management. Physician states diagnosis; patient must obtain treatment through local system.
Tier 4	Clinical guidance only (authorization pending)	Symptom-based guidance Urgency assessment Referral recommendation only No formal diagnosis	Used in jurisdictions where government authorization is in progress but not yet granted. Default tier for new markets.

		No treatment No prescribing	
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Note: Tier 4 is not a permanent state. It is a launch-safe default for markets where regulatory clearance is in progress. The platform must have a defined upgrade pathway from Tier 4 to higher tiers as authorizations are obtained.

2.3 Country Permission Matrix

The following table reflects the current known or anticipated permission status for priority launch markets. Tiers are assigned by Legal and Regulatory Affairs and stored in the Jurisdiction Registry (Section 6.3). This table is updated as authorizations are obtained.

Country / Region	Current Tier	Authorization Status	Notes & Constraints
India	Tier 2	In Progress	Telemedicine Practice Guidelines (2020) permit diagnosis via telemedicine. Prescribing of certain drug schedules restricted. State-level variation exists.
Rwanda	Tier 2	In Progress	Progressive digital health framework. MOH partnership recommended. Pharmacy integration [TBD].
Kenya	Tier 3	In Progress	Kenya Health Act permits licensed practitioners via telemedicine. Treatment management tier upgrade pending formal MOH agreement.
Nigeria	Tier 3	In Progress	MDCN Telemedicine Guidelines (2022) allow diagnosis. Prescribing authorization requires additional licensing step.
Brazil	Tier 3	In Progress	CFM Resolution 2.314/2022 permits telemedicine diagnosis. Prescribing allowed with digital signature infrastructure. LGPD data residency strictly enforced.
United States	Tier 2 (state-dependent)	In Progress	IMLC states: Tier 2 achievable. Non-IMLC states: Tier 3 or 4 until per-state licensure obtained. Ryan Haight Act limits controlled substance prescribing via telemedicine nationwide.
European Union	Tier 3	In Progress	EU Cross-Border Healthcare Directive permits telemedicine. Prescribing across member-state borders requires additional national authorization. GDPR data residency strictly enforced.
Mexico	Tier 3	[TBD]	NOM-004-SSA3-2012 governs clinical records. Formal telemedicine diagnostic

			authorization framework under development.
Philippines	Tier 4	[TBD]	Telehealth Act (RA 11223) exists but implementation regulations still evolving. Default Tier 4 until MOH clearance.
Default (all others)	Tier 4	Pending	Any jurisdiction without an explicit authorization entry defaults to Tier 4 (guidance only). Tier upgrade requires legal clearance and entry in the Jurisdiction Registry.

Constraint: This table is not a legal opinion. Each row must be validated by a licensed attorney in the relevant jurisdiction before the platform operates there. The tier assignments above are working estimates for engineering planning purposes only.

2.4 Physician Capability Profile

Each physician on the platform has a Capability Profile — a structured record that stores their jurisdiction-specific authorization flags. This profile is the source of truth for what a physician may do in a given session.

Profile Field	Type	Description
physician_id	UUID	Unique identifier
licensed_jurisdictions	Array<jurisdiction_id>	All jurisdictions where the physician holds an active, verified license
active_tier_per_jurisdiction	Map<jurisdiction_id, Tier>	Effective capability tier for each licensed jurisdiction — minimum of (country tier, physician license scope)
can_prescribe	Map<jurisdiction_id, boolean>	True only where both country Tier 1 is active AND physician holds prescribing rights in that jurisdiction
can_initiate_referral	Map<jurisdiction_id, boolean>	True for Tier 1 and Tier 2 jurisdictions
can_diagnose	Map<jurisdiction_id, boolean>	True for Tier 1, 2, and 3 jurisdictions
specialty	Array<string>	Clinical specialties — used for smart routing of complex cases
license_expiry_per_jurisdiction	Map<jurisdiction_id, date>	Licenses with upcoming expiry are flagged for renewal; expired licenses are auto-removed from routing eligibility
session_language_proficiency	Array<language_code>	Languages physician can conduct sessions in without translation support

3. Session Capability Disclosure — Interface Requirement

3.1 Requirement

At the start of every session — live video, audio call, or async video review — both the patient and the physician must be shown a Capability Disclosure Card. This card explicitly states what the attending physician is and is not authorized to do in the patient's jurisdiction for this session.

This is a non-negotiable interface requirement. It must appear before any clinical exchange begins and must be acknowledged by the patient with an explicit confirmation tap or click.

3.2 Patient-Facing Capability Disclosure Card

The following information must be displayed to the patient before the session begins. The language must match the patient's selected language (translated in real time if necessary).

Example Disclosure Card (Tier 2 — Kenya, Dr. Amara Osei)

Your doctor today:

Dr. Amara Osei — General Practitioner, licensed in Kenya

In your region, Dr. Osei is authorized to:

- ✓ Examine your symptoms and give you a formal diagnosis
- ✓ Recommend a treatment plan and home care steps
- ✓ Refer you to a specialist or in-person facility

In your region, Dr. Osei is NOT authorized to:

- ✗ Prescribe medication directly — you will need to obtain a prescription from a local provider or pharmacy partner
- ✗ Order laboratory tests or imaging directly through this platform

This session is not a substitute for emergency care. If your condition is life-threatening, please contact emergency services immediately.

[I understand and wish to continue] ■ Confirm

3.3 Physician-Facing Capability Reminder

The physician's session interface must also display their active capability tier for the patient's jurisdiction. This appears as a persistent status bar throughout the session, not only at the start.

Physician Session Status Bar (persistent, top of session UI)

Patient jurisdiction: Kenya | Your tier: **Tier 2 — Diagnosis + Treatment**

✓ Diagnose ✓ Treat ✗ Prescribe ✓ Refer

License verified: Kenya Medical Practitioners Board — expires 31 Dec 2026

3.4 Disclosure Rules by Tier

Tier	Session Disclosure Label	Patient confirmation required?	Physician status bar color
Tier 1	Full diagnosis, treatment & prescribing authorized	Yes — before session start	Green
Tier 2	Full diagnosis & treatment authorized — prescribing not available in your region	Yes — before session start	Teal
Tier 3	Diagnosis authorized — treatment management handled by local provider	Yes — before session start	Blue
Tier 4	Clinical guidance only — formal diagnosis not authorized in your region	Yes — before session start, with additional emphasis on limitations	Amber — warning tone

3.5 Async Modality Disclosure

For asynchronous video submissions, the capability disclosure is delivered at two points:

- At submission time — patient sees the disclosure before confirming the upload, informing them what the reviewing physician will be authorized to provide in their jurisdiction.
- In the guidance response — the written response from the physician includes a standardized footer that restates their authorization level and any capability limitations that affect the guidance provided.

[TBD] Disclosure footer template for async responses to be developed by CMO and legal counsel.

4. System Overview

4.1 Core Modalities

Modality	Description	Capability Tier Applies
Live Video	Real-time video call with translation and annotation tools. Physician capability status bar active throughout.	Yes — tier governs what physician may state, document, or prescribe during session
Audio Call	Real-time voice-only session for low-bandwidth patients or those without video capability. Physician verbal disclosures must mirror the written capability card.	Yes — physician must verbally state limitations at session open
Async Video	Patient submits recorded symptom video. Physician reviews and responds within SLA window.	Yes — tier governs guidance response scope; stated in response footer

4.2 User Journey with Capability Gate

1. Patient completes intake questionnaire (body map, severity, history).
2. AI triage assigns urgency and routes to modality. Jurisdiction is detected or declared.
3. Platform looks up patient jurisdiction in Jurisdiction Registry — retrieves current tier.
4. Platform identifies eligible physicians (licensed in jurisdiction, correct specialty, available).
5. Patient is shown Capability Disclosure Card for their jurisdiction and the assigned physician.
6. Patient acknowledges disclosure — session begins (live/audio) or submission proceeds (async).
7. Physician session interface shows persistent capability status bar.
8. Physician conducts session within authorized tier scope.
9. Physician issues guidance response. Response template enforces tier-appropriate language.
10. Capability tier and physician ID are recorded in the consultation record for audit purposes.

4.3 Non-Goals (Out of Scope v1.0)

- Controlled substance prescribing — excluded from all tiers permanently
- Diagnostic imaging upload and radiology review
- Wearable device data integration
- Insurance billing and claims processing

- FHIR health record portability — [TBD] Phase 3

5. System Architecture

5.1 Key Architectural Additions (v0.2)

The shift to a diagnosis-permission model introduces two new architectural components that did not exist in v0.1:

- Jurisdiction Registry Service — the authoritative source of tier assignments per country. All routing and disclosure logic reads from this service.
- Capability Enforcement Layer — a middleware component that intercepts physician response submissions and validates that the response content does not exceed the authorized tier for the patient's jurisdiction. Flags out-of-scope content for physician review before delivery.

5.2 Technology Stack

Layer	Candidate / Decision	Notes
Frontend (Patient)	[TBD]	Must support low-end Android, 3G; disclosure card must render before session initiation
Frontend (Physician)	[TBD]	Persistent capability status bar required in session view
Backend API	[TBD]	REST or GraphQL TBD
Jurisdiction Registry	Internal service — PostgreSQL backed	Source of truth for tier per country; versioned; audit-logged on every change
Capability Enforcement Layer	Custom middleware	Intercepts physician response; validates against tier before delivery
Real-time Video	[TBD]	HIPAA-eligible vendor required
Audio Call	[TBD]	Must support 64kbps minimum; verbal disclosure flow required
Async Video Storage	[TBD]	Geographic data residency per patient jurisdiction
AI Triage Engine	[TBD]	SaMD regulatory review required; outputs urgency only
Translation	[TBD]	Medical glossary layer required; disclosure card translated in real time
Database	[TBD]	Per-region instance for patient data; central instance for physician profiles
Auth & Identity	[TBD]	Physician MFA required; license verification integration required

6. Core Modules

6.1 Jurisdiction Registry Service

The Jurisdiction Registry is the single source of truth for all capability tier assignments. It is a versioned, audit-logged service that the routing engine, disclosure card generator, response template engine, and capability enforcement layer all read from.

6.1.1 Registry Record Structure

jurisdiction_id	ISO 3166-1 alpha-2 country code (e.g., KE, IN, BR)
country_name	Human-readable country name
current_tier	Tier 1 / 2 / 3 / 4
tier_effective_date	Date the current tier was activated
authorization_document_ref	Reference to legal authorization document stored in compliance vault
prescribing_permitted	Boolean — Tier 1 only; may be further restricted by drug schedule
referral_initiation_permitted	Boolean — Tier 1 and 2
lab_ordering_permitted	Boolean — [TBD] for all jurisdictions currently
notes	Free text — any specific country constraints or caveats
last_reviewed_by	Legal/Regulatory Affairs officer ID
next_review_date	Scheduled legal review date for this jurisdiction's tier status

6.1.2 Access Rules

- Read access: All platform services (routing, disclosure, response templates, enforcement layer)
- Write access: Legal and Regulatory Affairs team only, with dual-approval workflow
- Every tier change triggers an audit log entry and notifies the CMO and General Counsel
- Physicians currently serving patients in an affected jurisdiction are notified of tier changes before their next session

6.2 Intake & Triage Module

6.2.1 Inputs

- Body map — patient selects anatomical region(s) of concern
- Pain/severity scale — numeric 1–10 or categorical
- Symptom duration and onset (sudden vs. gradual)

- Basic medical history flags — allergies, chronic conditions, medications
- Age and biological sex

6.2.2 Triage Outputs

- Urgency level: Emergency / High / Medium / Low
- Routing decision: Live Video / Audio Call / Async Video / Emergency Services
- Patient jurisdiction lookup — tier retrieved from Jurisdiction Registry
- Eligible physician pool — filtered by jurisdiction eligibility AND capability tier

Note: All triage outputs urgency and routing only. No condition names. No diagnostic language. See Section 9.

6.2.3 Hard Emergency Exclusions (Non-Overridable)

- Chest pain or pressure
- Stroke indicators (facial drooping, arm weakness, speech difficulty)
- Severe difficulty breathing
- Major trauma or injury
- Loss of consciousness
- Suicidal ideation or self-harm

[TBD] Full exclusion list to be finalized by CMO.

6.3 Live Video Modality

6.3.1 Session Start Sequence

11. Physician assigned from eligible pool — capability profile loaded.
12. Capability Disclosure Card generated for patient (tier-specific, translated).
13. Patient acknowledges disclosure — session initiates.
14. Physician capability status bar rendered at top of session UI.
15. Translation layer activates.

6.3.2 Response Enforcement

- The Capability Enforcement Layer scans the physician's documented session notes and response before they are finalized.
- If the response contains prescribing language in a non-Tier-1 jurisdiction, it is flagged and the physician is prompted to revise.
- Physician may override a flag with a documented clinical rationale — this is logged for audit purposes.

[TBD] Specific language rules and flagging logic for the Capability Enforcement Layer to be defined by CMO and legal counsel.

6.4 Audio Call Modality

Note: Full feature scope under active evaluation. Sections below reflect current planning assumptions.

6.4.1 Verbal Capability Disclosure

Because the audio modality has no visual interface during the call itself, the physician must verbally deliver a standardized capability disclosure at the start of every call. A scripted disclosure prompt is displayed on the physician's interface before the call connects.

Example verbal disclosure prompt (physician sees this before call connects):

"Before we begin, I am required to let you know what I can help you with today. In your region, I am authorized to examine your symptoms, give you a formal diagnosis, and recommend a treatment plan. I am not authorized to prescribe medication through this platform — if medication is needed, I will direct you to a local provider. Do you understand and wish to continue?"

6.4.2 Open Technical Questions

[TBD] Audio call queue — shared with live video queue or dedicated? Impact on physician scheduling model.

[TBD] Speech-to-speech translation quality validation for real-time medical conversation.

[TBD] Automatic fallback trigger threshold from live video to audio call on bandwidth degradation.

[TBD] Minimum audio quality threshold for translation layer reliability.

6.5 Async Video Modality

6.5.1 Patient Flow

16. Patient triaged to async — receives confirmation with expected review window.
17. Capability Disclosure Card shown before submission — patient acknowledges.
18. Patient records symptom video guided by in-app prompts.
19. Video submitted with intake data attached.
20. AI processes submission — generates annotation draft and urgency signal.
21. Case enters physician review queue — tier, urgency, and annotation draft visible.
22. Physician reviews within SLA. Capability Enforcement Layer validates response scope.
23. Pre-response check-in sent to patient: status confirmation required.
24. Guidance response delivered with tier disclosure footer.

6.5.2 Submission Constraints

- Maximum video duration: [TBD] — likely 2–5 minutes
- Submission expiry: 6–12 hours — stale submissions require patient re-confirmation
- Emergency symptom profiles: excluded from async — routed to live queue or emergency services
- Audio-only async: permitted as fallback for patients unable to record video

6.6 Translation Layer

Text-to-text (UI, forms, responses)	All modalities — vendor [TBD]
Speech-to-text + captions (live)	Live video and audio call — vendor [TBD]
Speech-to-speech (real-time voice)	Audio call — [TBD] quality validation required
Async transcript translation	Applied post-submission — vendor [TBD]
Disclosure card translation	Real-time at session start — must precede clinical exchange

Constraint: Disclosure cards must be translated before session start. A session cannot begin if the disclosure card translation fails or is incomplete.

7. Data Architecture

7.1 Core Data Entities

Entity	Key Attributes	Notes
Patient	ID, region, language, device profile, consent flags	PII — encrypted; regional data residency enforced
Intake Record	Patient ID, timestamp, body map, severity, history	Attached to every consultation regardless of modality
Consultation	ID, modality, patient ID, physician ID, jurisdiction_id, tier_at_time_of_session, status, SLA timestamps	tier_at_time_of_session is immutable after session close — preserves audit record of authorization at time of care
Disclosure Acknowledgment	Consultation ID, patient ID, tier shown, timestamp, language, acknowledgment flag	Required for every session — stored indefinitely for legal audit
Video Submission	Consultation ID, file reference, encryption key ID, expiry timestamp	Async only; regional bucket; auto-deleted per retention policy
Audio Session Record	Consultation ID, recording reference (if consented), transcript, translation, verbal_disclosure_confirmed flag	Physician must confirm verbal disclosure was delivered before call connects
AI Annotation	Consultation ID, body map annotations, urgency signal, draft flag, physician override status	Draft until physician confirms
Physician	ID, licensed_jurisdictions, active_tier_per_jurisdiction, can_prescribe, specialty, license_expiry_per_jurisdiction	Jurisdiction array drives all routing eligibility
Guidance Response	Consultation ID, physician ID, disposition type, response text, tier_footer, translation, timestamp	tier_footer auto-appended — physician cannot remove it
Consent Record	Patient ID, consent type, version, timestamp	Granular per consent type; stored indefinitely
Jurisdiction Registry Record	jurisdiction_id, current_tier, effective_date, authorization_doc_ref, last_reviewed_by	Write-protected; dual-approval required for changes

7.2 Immutable Audit Fields

The following fields must be immutable once written. They form the legal audit trail and cannot be modified or deleted, including by administrators:

- tier_at_time_of_session on the Consultation record
- Disclosure Acknowledgment records
- Jurisdiction Registry change log entries
- Physician Capability Profile change history
- Consent Records

8. Security & Compliance

8.1 Encryption

- All data in transit: TLS 1.3+
- All data at rest: AES-256 with per-patient key isolation
- Encryption key management: [TBD] — AWS KMS / HashiCorp Vault / per-region key service

8.2 Authentication & Access Control

- Patients: [TBD] — email/SMS OTP, social login, or device biometric
- Physicians: Multi-factor authentication required; license verification integration
- RBAC enforced at API layer — physician dashboard scoped to licensed jurisdictions only
- Jurisdiction Registry: write access restricted to Legal/Regulatory team with dual approval

8.3 Compliance Frameworks

Framework	Jurisdiction	Key Requirements
HIPAA	United States	BAAs with all vendors; minimum necessary access; breach notification 60 days
GDPR	European Union	[TBD]
LGPD	Brazil	Data localization; consent; national authority notification
[TBD]	Kenya, Rwanda, India, Nigeria	[TBD]

8.4 Consent Architecture

- Video/audio storage consent — separate from session participation
- AI processing consent — explicit, separate
- Translation processing consent — third-party data exposure disclosed
- Research use consent — optional
- Disclosure acknowledgment — mandatory, logged per session

9. Regulatory & Legal Constraints

9.1 Diagnosis Authorization

- Diagnosis is only permitted in jurisdictions with an active Tier 1, 2, or 3 designation in the Jurisdiction Registry.
- The authorization document for each jurisdiction is stored in the compliance vault and linked to the Registry record.
- If authorization lapses or is revoked, the jurisdiction automatically downgrades to Tier 4 and all active sessions in that jurisdiction are notified.

[TBD] Lapsed authorization alert workflow and physician notification protocol to be defined by Legal and Operations.

9.2 AI as Software as a Medical Device (SaMD)

- AI triage and annotation tools are candidate SaMD under FDA 21 CFR Part 882 and EU MDR Annex VIII Rule 11.
- AI must remain decision support only — all outputs are drafts requiring physician confirmation.
- FDA Breakthrough Device Program application to be filed before US launch.
- EU MDR fast-track pathway to be pursued for EU launch.

[TBD] Regulatory affairs engagement timeline and responsible party TBD.

9.3 Capability Enforcement

- The Capability Enforcement Layer is a compliance control, not a clinical one. It catches obvious tier violations in physician responses before delivery.
- It does not replace physician judgment — physicians may override a flag with documented rationale.
- All flag events (flagged, overridden, corrected) are logged and reviewed monthly by the CMO.

9.4 Async Liability Controls

- Submission expiry: 6–12 hours; stale submissions require patient re-confirmation before physician response.
- Pre-response check-in: automated patient status ping before guidance delivery.
- Emergency escalation: automatic if patient reports worsening or does not respond within 2 hours.
- Hard emergency exclusions: hardcoded; non-overridable (Section 6.2.3).

10. Non-Functional Requirements

Category	Requirement	Target / Notes
Performance	Live video session latency	< 150ms round-trip
Performance	Disclosure card render + translation time	< 2 seconds — session cannot start until complete
Performance	Triage engine response time	< 3 seconds from intake submission
Performance	Jurisdiction Registry lookup	< 100ms — must not add perceptible delay to session start
Availability	Platform uptime	99.5% target
Availability	Jurisdiction Registry uptime	99.9% — higher SLA required as it gates all sessions
Scalability	Concurrent live sessions	[TBD]
Accessibility	WCAG compliance	WCAG 2.1 AA minimum — disclosure card must meet same standard
Device support	Minimum patient device	[TBD]
Localization	Disclosure card languages	All launch languages — RTL layout required
Audit	Disclosure acknowledgment logging	100% capture rate — any gap is a compliance failure

11. Open Items & TBD Registry

#	Open Item	Owner	Target Milestone
1	Technology stack selection — frontend, video, audio vendors	Engineering	Phase 1 — month 2
2	Audio call full feature scope and queue architecture	Product + Clinical	Phase 1 — month 3
3	Speech-to-speech translation quality validation for audio call	Clinical + Vendor	Phase 1 — month 4
4	Jurisdiction tier assignments validated by licensed attorneys per country	Legal	Phase 1 — month 3
5	Authorization documents obtained for Tier 2+ markets (India, Rwanda, Kenya priority)	Regulatory Affairs	Phase 1 — month 4
6	Capability Enforcement Layer — specific flagging rules and physician override protocol	CMO + Legal	Phase 1 — month 4
7	Disclosure card template per tier — final language reviewed by legal counsel	Legal + CMO	Phase 1 — month 3
8	Verbal disclosure script for audio call modality	CMO	Phase 1 — month 3
9	Async response tier footer template	CMO + Legal	Phase 1 — month 3
10	Translation vendor selection and medical glossary development	Clinical + Vendor	Phase 1 — month 3
11	AI triage model architecture — custom vs. third-party	Engineering + Clinical	Phase 1 — month 4
12	SaMD regulatory filing — FDA Breakthrough Device and EU MDR fast-track	Regulatory Affairs	Phase 1 — month 2
13	Data residency infrastructure mapping per jurisdiction	Engineering + Legal	Phase 1 — month 3
14	Lapsed authorization alert and session protocol	Legal + Operations	Phase 1 — month 4
15	Lab ordering and imaging — feasibility per Tier 1 jurisdictions	CMO + Legal	Phase 2
16	e-Prescribing pharmacy integration — Tier 1 jurisdictions	Engineering + Legal	Phase 2
17	FHIR health record portability	Engineering	Phase 3+

12. Appendix

A. Glossary

Capability Tier	The jurisdiction-specific scope of clinical action authorized for the platform in a given country. Ranges from Tier 1 (full diagnosis + prescribing) to Tier 4 (guidance only).
Jurisdiction Registry	Internal versioned service storing the current tier assignment for every country where the platform operates.
Capability Enforcement Layer	Middleware that validates physician response content against the active tier before delivery to the patient.
Disclosure Card	Patient-facing UI element shown before every session that states what the attending physician is and is not authorized to do in the patient's jurisdiction.
SaMD	Software as a Medical Device — software that influences clinical decisions; subject to FDA/MDR regulation.
SLA	Service Level Agreement — defined maximum response window for async submissions.
IMLC	Interstate Medical Licensure Compact — US multi-state physician licensing framework.
HIPAA	Health Insurance Portability and Accountability Act — US health data privacy law.
GDPR	General Data Protection Regulation — EU data privacy law.
LGPD	Lei Geral de Proteção de Dados — Brazilian data privacy law.
CMO	Chief Medical Officer.
RBAC	Role-Based Access Control.
TLS	Transport Layer Security — encryption protocol for data in transit.
Tier 1	Full diagnosis, treatment recommendation, and prescribing authorized.
Tier 2	Diagnosis and treatment authorized; prescribing not permitted.
Tier 3	Diagnosis authorized; treatment management handled by local provider.
Tier 4	Clinical guidance only; formal diagnosis not yet authorized. Default for new markets.

B. Reference Documents

- Open Access Telehealth Platform — Project Summary (.txt)
- SDD v0.1 — Initial Outline Draft

- WHO Global Strategy on Digital Health 2020–2025
- FDA Guidance: Software as a Medical Device (SaMD) — Clinical Evaluation
- EU Medical Device Regulation (MDR) 2017/745 — Annex VIII Rule 11
- India Telemedicine Practice Guidelines — MCI/NMC (2020)
- CFM Resolution 2.314/2022 — Brazil Telemedicine
- MDCN Telemedicine Guidelines — Nigeria (2022)
- Kenya Health Act — Telemedicine Provisions
- HHS HIPAA Security Rule (45 CFR Parts 160 and 164)
- GDPR Article 9 — Special Categories of Personal Data
- Interstate Medical Licensure Compact (IMLC) — imlcc.org

C. Document Revision History

Version	Date	Author	Summary
0.1	Prior session	[TBD] — Project Lead	Initial outline — guidance-only model, structure and scope
0.2	March 28, 2026	[TBD] — Project Lead	Revised assumption: government authorization for diagnosis. Added Jurisdictional Permission Model, Capability Tiers, Country Matrix, Session Disclosure Interface, Capability Enforcement Layer.